

Topic 28 -- Taylor and Maclaurin Series

We've seen a few situations where we can create a power series representation of a function by careful "restructuring" of a known function and/or with the use of differentiation and integration.

Is there a more general approach?

Suppose that a function has a power series representation:

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$$

Conclusions:

If f has a power series representation (expansion) at a , that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n \quad |x - a| < R$$

Then, its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}$$

So, the expansion at $x = a$ is given by:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

This is called the Taylor series of the function f at a .

If we take the Taylor series and let $a = 0$,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

This is called the Maclaurin series of f .

Examples

Determine a series representation for $f(x) = e^x$ about $x = 0$. Find the associated radius of convergence.

Determine a series representation for $f(x) = \cos x$ about $x = 0$. Find the associated radius of convergence.

Determine a series representation for $f(x) = \sin x$ about $x = 0$. Find the associated radius of convergence.

Determine a series representation for $f(x) = \ln(1 + x)$ about $x = 0$. Find the associated radius of convergence.

Determine a series representation for $f(x) = \frac{1}{x}$ about $x = 3$. Find the associated radius of convergence.

Determine the first three terms of a series representation for $f(x) = \sqrt{x}$ about $x = 9$. Use it to approximate $\sqrt{10}$

Some important series representations:

Function and Representation	Radius
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$	$R = 1$
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$R = \infty$
$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	$R = \infty$
$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	$R = \infty$
$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$	$R = 1$
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$R = 1$
$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$	$R = 1$

Examples:

Determine a series representation for $f(x) = x^2 \ln(1 + x^3)$

Find a degree-4 polynomial that may be used to approximate $f(x) = \frac{x^2}{\sqrt{9+x}}$

If $i^2 = -1$, use series to show that $e^{ix} = \cos x + i \sin x$.

Taylor Remainder and Estimation

If $|f^{(n+1)}(x)| \leq M$ for $|x - a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$$

Determine the number of terms of the series representation of $f(x) = e^x$ about $x = 0$ to approximate $e^{0.1}$ accurate to within 0.00001.

Calculate the upper bound of the error of using the degree-3 Maclaurin approximation of $\sin 1$ in two ways: using the Taylor Remainder Estimation and the Alternating Series Estimation. Which gives a better error bound?

Let $f(x) = x^3 - 3x + 7$. Write $f(x)$ as a Taylor series about $a = 1$. Use the Taylor series to find $f(1.1)$.

Approximate the integral to an accuracy of 0.001.

$$\int_0^1 \sin(x^2) dx$$